

# INHOMOGENEITIES IN THE UNIVERSE

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NTU – 21st October

# Inhomogeneities – What are we talking about?

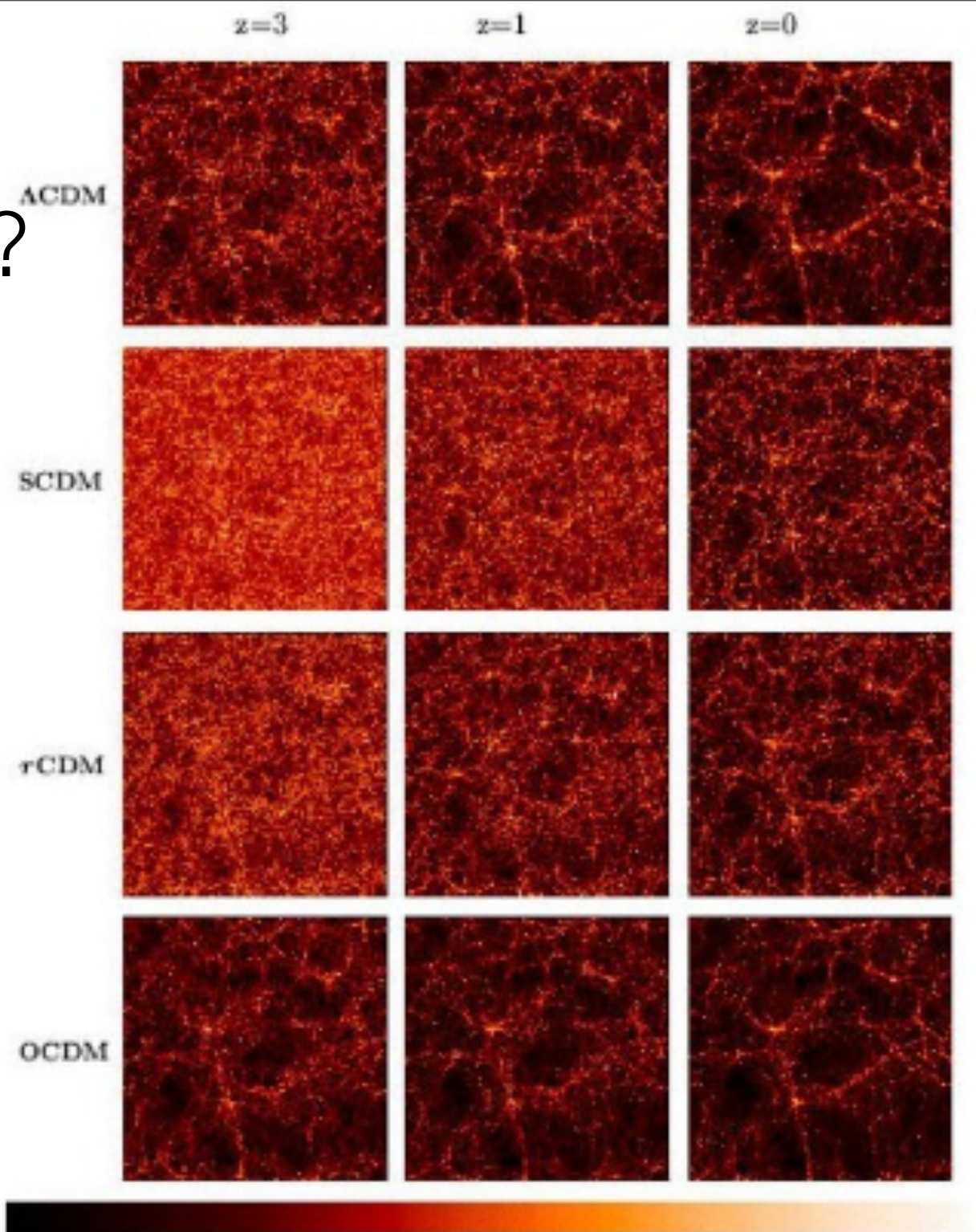
- Work with the density contrast

$$\delta(\mathbf{x}, t) = \frac{\rho(\mathbf{x}, t) - \rho_{\text{cr}}(t)}{\rho_{\text{cr}}(t)},$$

- Do this in Fourier space  
Cartoon equation:

$$\ddot{\delta} + (\text{pressure} - \text{gravity})\delta = 0,$$

- How do we put this in the framework of GR?  
Can we find an explanation for the highly nonlinear structure we observe today?



The VIRGO Collaboration 1996

nonlinear structure evolution, simulated in different cosmologies  
(Virgo collaboration)

# To explain matter perturbations we only invoke perturbations in metric, matter and radiation!

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- First order perturbations in the metric introduce the potential  $\Phi$
- Working with dark matter particles gives the quantities:  
 $\delta$ , the density of particles, and  
 $v$ , the velocity of the particles
- For radiation we use two more variables (only that is! ):  
 $\Theta_{r,0}$  the monopole moment of radiation  
 $\Theta_{r,1}$  the dipole moment of radiation
- We now have 5 variables in k-space!

# One comment on the metric:

The metric now has  $\Phi$  and  $\Psi$ , where we set  $\Phi = -\Psi$

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- Known form of the metric (Friedmann Robertson Walker)

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & a^2(t) & 0 & 0 \\ 0 & 0 & a^2(t) & 0 \\ 0 & 0 & 0 & a^2(t) \end{pmatrix}$$

- Perturbation of the metric in first order:

Simply add factors of  $(1+2\Psi(x,t))$  and  $(1+2\Phi(x,t))$  to find

$$g_{\mu\nu} = \begin{pmatrix} -1(1+2\Psi) & 0 & 0 & 0 \\ 0 & a^2(t)(1+2\Phi) & 0 & 0 \\ 0 & 0 & a^2(t)(1+2\Phi) & 0 \\ 0 & 0 & 0 & a^2(t)(1+2\Phi) \end{pmatrix}$$

We will be quite interested in  $\Phi$ !

The full Boltzmann-Einstein system looks much more complicated than it is: Doesn't even have Compton scattering!

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- Boltzmann Einstein system

$$\begin{aligned}\dot{\Theta}_{r,0} + k\Theta_{r,1} &= -\dot{\Phi} \\ \dot{\Theta}_{r,1} + \frac{k}{3}\Theta_{r,0} &= \frac{-k}{3}\Phi \\ \dot{\delta} + ikv &= -3\dot{\Phi} \\ \dot{v} + \frac{\dot{a}}{a}v &= ik\Phi,\end{aligned}$$

- One more equation

$$k^2\Phi + 3\frac{\dot{a}}{a}\left(\dot{\Phi} + \frac{\dot{a}}{a}\Phi\right) = 4\pi Ga^2(\rho_{\text{dm}}\delta + 4\rho_r\Theta_{r,0}).$$

# The most straightforward thing to do: A numerical simulation. So let's do that!

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- Here we use a integration method in C++ from “Numerical Recipes” William Press, Saul Teukolsky, William Vetterling, and Brian Flannery. *Numerical Recipes in C*. Cambridge University Press, Cambridge, UK, 3rd edition, 2007.
- There is ONE definition to keep in mind
- There are THREE resulting plots to keep in mind:
  - potential  $\Phi$
  - growth factor  $D_1$
  - power spectrum

## The Boltzmann-Einstein system

The next structure is used to integrate the full Boltzmann-Einstein system given by the equations (2.4) and the following on page 18.

```
struct cosmo_full{  
2  Doub k, eta , a_dot, phi_dot;  
  cosmo_full(Doub kk): k(kk) {}  
4  void operator() (const Doub logeta , VecDoub_I &y , VecDoub_O &  
    dydx){
```

# Evolution of the potential $\Phi$ shows the effect of the transfer function up to a factor of 9/10.

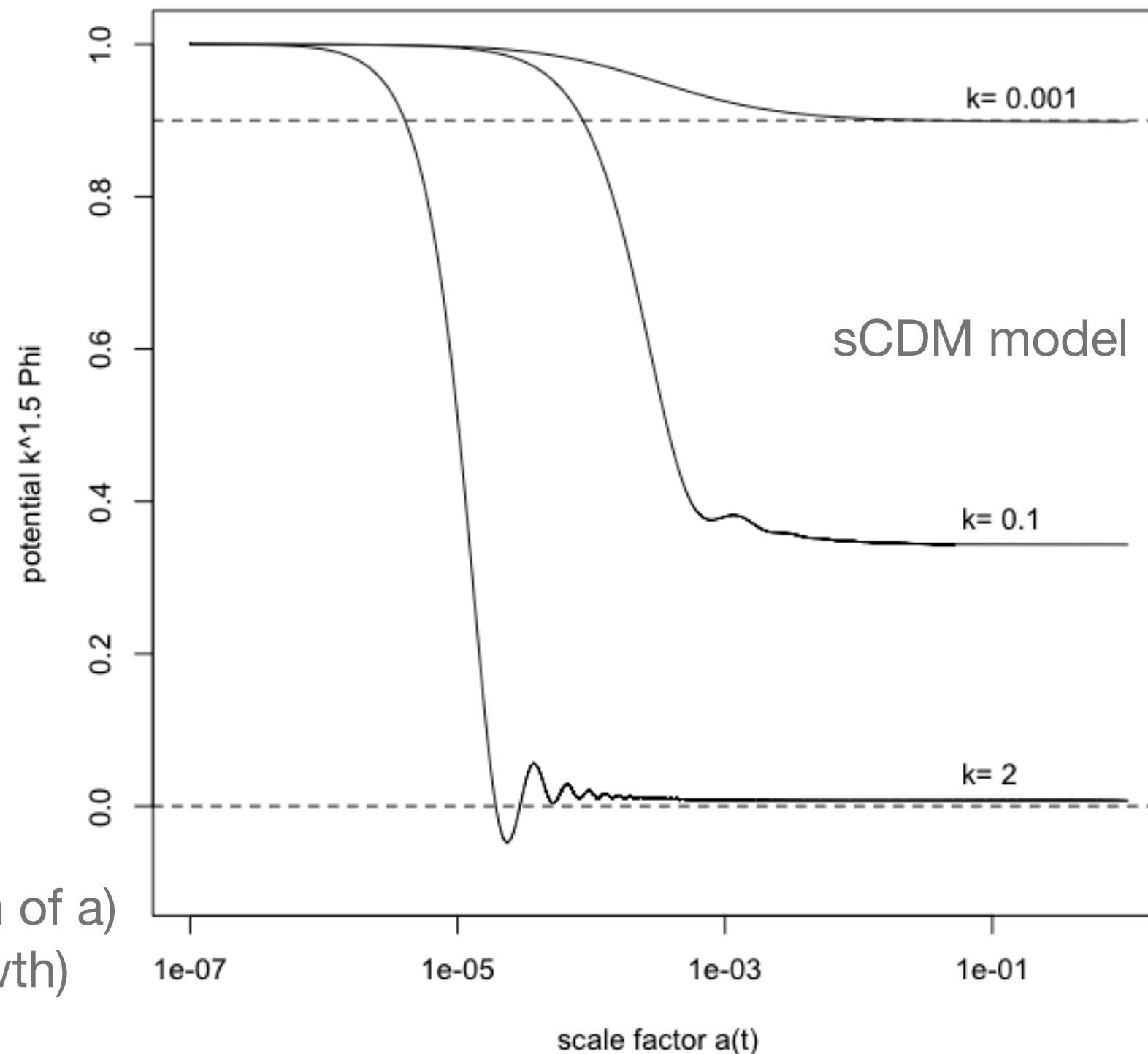
- definition:

$$\Phi(\mathbf{k}, a) = \frac{9}{10} \Phi_P(\mathbf{k}) T(k) \frac{D_1(a)}{a}$$

primordial value

transfer function  
(function of  $k$ )

growth factor (function of  $a$ )  
(normed by linear growth)

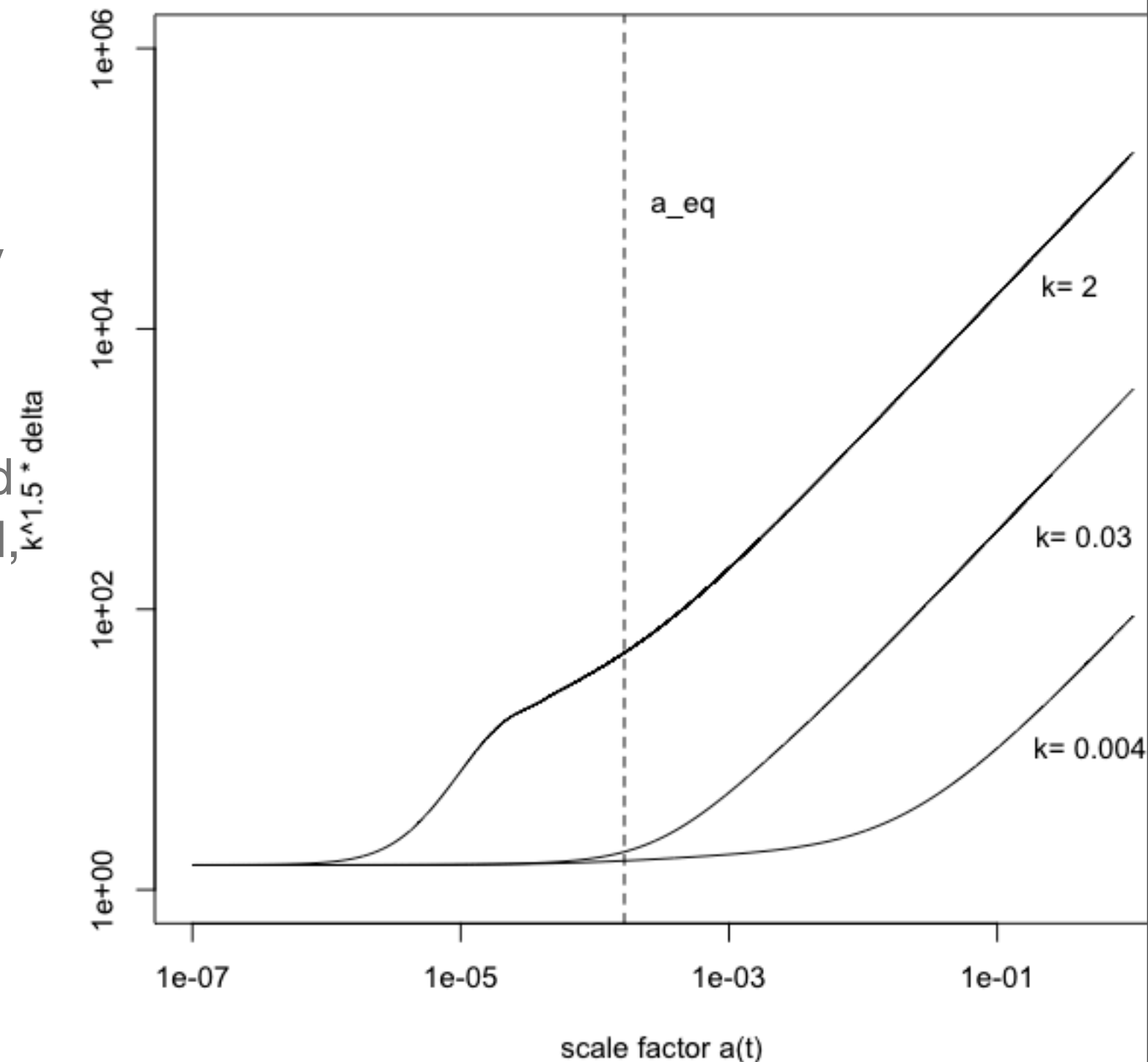


Substantial drop for small scale modes. At late times the potential stays constant.



# Where the growth factor got its name from...

- Although  $\Phi$  const at late times, the inhomogeneities grow!
- In **matter** dominated phase they grow linearly!
- However, in **radiation** dominated phase the growth is suppressed, but still present.



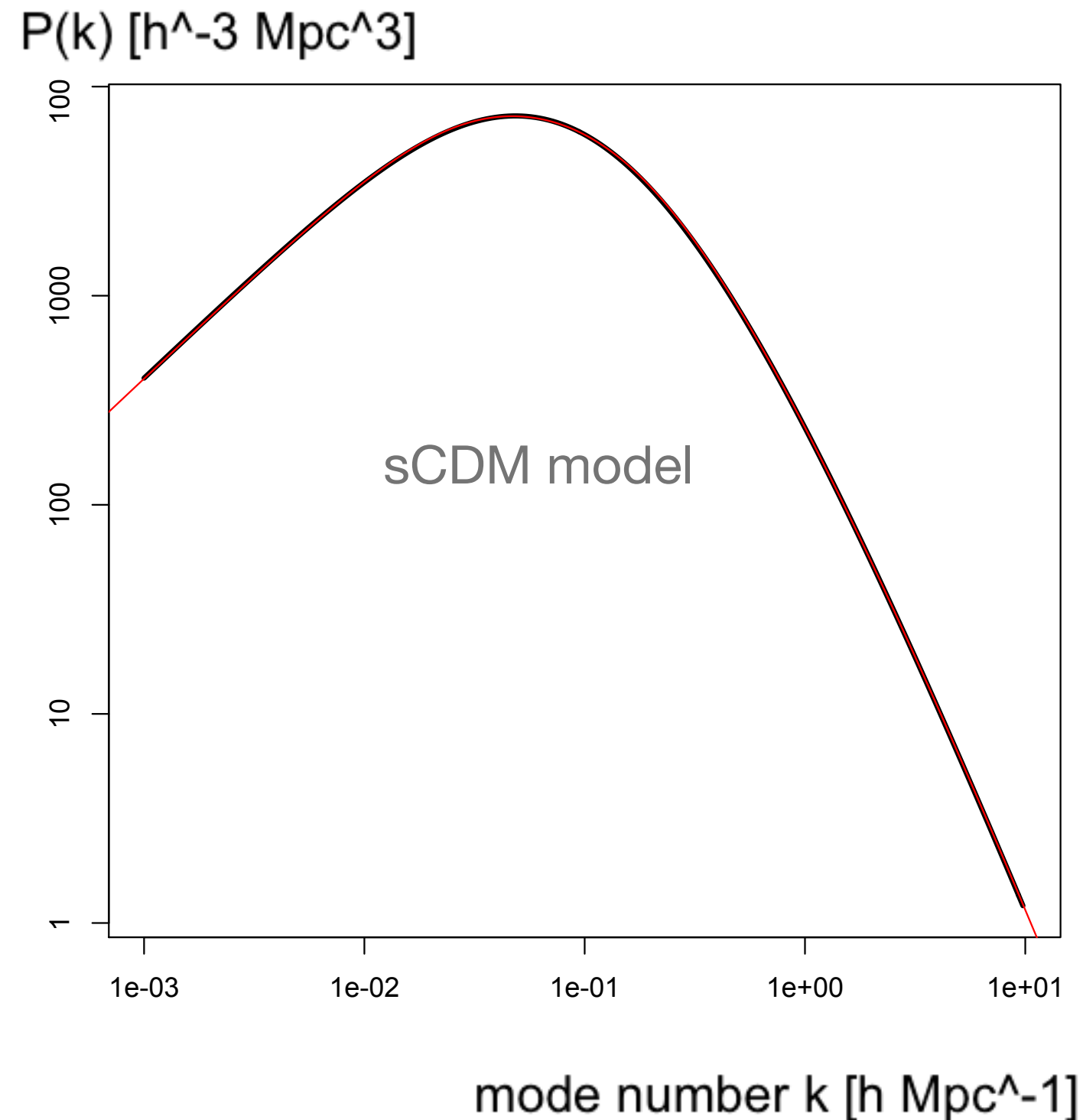


# The power spectrum has a turnover point.

- Large scale modes enter at late times, they evolve without obstruction;

Small scales entered during radiation domination: highly suppressed.

- The turnover point is very sensitive to the model we used:  
(here sCDM and  $\Lambda$ CDM)
- and it IS observationally accessible.



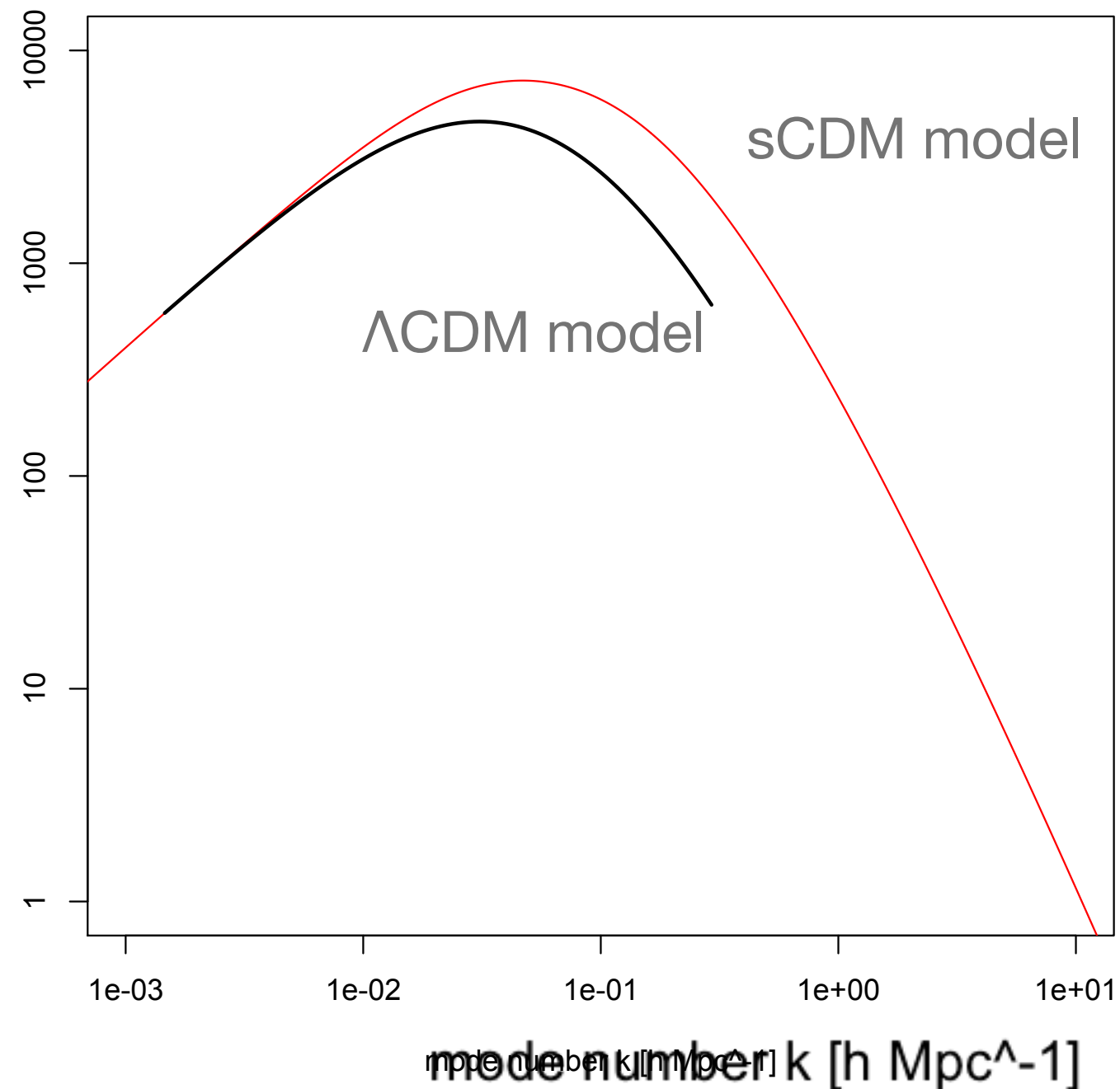
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(here sCDM and  $\Lambda$ CDM)
- and it IS observationally accessible:  
 $\Lambda$ CDM model favoured!

$P(k) [h^{-3} \text{ Mpc}^3]$



The power spectrum definitely is something people talk about!

# Outline

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- The turnover point in the power spectrum of the  $\Lambda$ CDM model can be seen in the simulation. What about the small scale modes not shown in the diagram?
- Have seen the model dependence of the power spectrum. We can ask the same question for the growth factor and transfer function.
- What about other models? How do dark energy models compare to the cosmological constant models presented here?
- Using this approach can we find constraints on the dark energy equation of state?